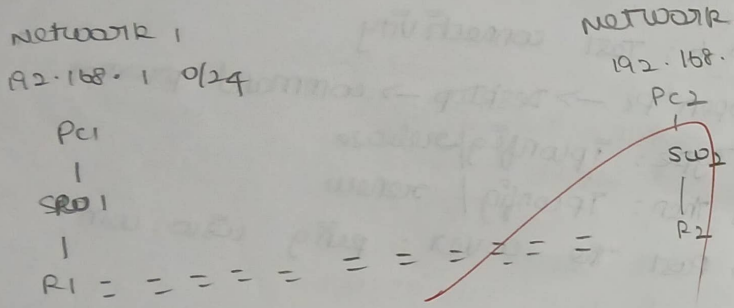


Pg 9

Experiment - 11

Routing at Network layer

a) Simulate static Routing Protocol configuration using cisco packet tracer

Aim: To configure and simulate static routing with a backup route using cisco packet tracer.

Requirements:
- cisco packet tracer
- 3 routers
- Pcs/ end devices
- Proper IP addresses
- Ethernet / serial connection

Router 1: Router 1 was configured with static routes to the 10.0.0.0/8 and 30.0.0.0/8 networks. A primary route with Administrative Distance 10 and a backup route with AD 20 were configured.

Router 2: Router 2 was configured with static routes to the 10.0.0.0/8 and 30.0.0.0/8 networks.

Checking ip route static: Successful ping replies confirmed connectivity between the source and destination networks

Deleting Static Route: After deleting the primary route, the backup route became active, demonstrating redundancy using floating static routes.

Result: Thus, static routing with primary and backup route was successfully configured and simulated using cisco packet tracer.

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1) Simulate RIP using cisco packet Tracer

Aim:

To simulate and configure RIP routing using cisco packet Tracer and verify connectivity between different networks

Observation:

* A network topology was created using Router 0, Router 1 and Router 2.
* Required IP addresses were assigned to router interfaces and PCs.
* The routing table was checked using show ip route.

Router 1: Router 1 was configured with routes to the 10.0.0.0/8 and 40.0.0.0/8 networks. The routes were verified using show ip route.

Router 2: Router 2 was configured with routes to 10.0.0.0/8 through Router 0 and 30.0.0.0/8 through Router 1.

Verification: connectivity was tested using ping and Tracert. successful replies confirmed that packets reached the destination networks.

Backup route verification: The main link between Router 0 and Router 1 was disconnected. the backup path through Router 2 was selected.

Result: Thus, RIP routing was successfully configured and simulated using cisco packet Tracer.